

四. 波动光学

1. 光的本质: 判断题

① 光是电磁波:

$$\text{光波} \begin{cases} \text{红外光: } \lambda > 760 \text{ nm} \\ \text{可见光: } 400 \text{ nm} < \lambda < 760 \text{ nm} \\ \text{紫外光: } \lambda < 400 \text{ nm} \end{cases}$$

光的颜色由频率决定, 光在不同介质中颜色保持不变(频率不变)。

② 光子:

光是一种具有波粒二象性的电磁波。

2. 光的相干性、光程与光程差: 选择题

一. 光的干涉现象:

发生干涉现象的两列光波的条件

- 振动频率相同
- 相位相同或相位差恒定
- 光矢量方向相同
- 两列光波光强差不能太大

获取相干光的办法

- (1) 分波面法: 杨式双缝干涉
- (2) 分振幅法: 薄膜干涉

二. 光程差:

光程: $L = nr$ 折射率 $\times$ 传播距离

$$\Delta L = L_1 - L_2$$

$$\Delta L = \begin{cases} \pm k \frac{\lambda}{2}, & k=0, 1, 2, \dots \text{干涉相长} \\ \pm (2k+1) \frac{\lambda}{2}, & k=0, 1, 2, \dots \text{干涉相消} \end{cases}$$

$$\Delta \phi = 2\pi \frac{\Delta L}{\lambda}$$

3. 光的干涉: 选择题

光的干涉

- ① 杨式双缝干涉: $\delta = d \sin \theta = d \tan \theta = \frac{d}{D} x = \begin{cases} \pm k \lambda, & \text{明纹} \\ \pm (2k+1) \frac{\lambda}{2}, & \text{暗纹} \end{cases}$
- ② 薄膜干涉
 - 等厚干涉: $\delta = 2nd(1 + \frac{\lambda}{2}) = \begin{cases} \pm k \lambda \\ \pm (2k+1) \frac{\lambda}{2} \end{cases}$
 - 等倾干涉: $\delta = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} (\pm \frac{\lambda}{2}) = \begin{cases} \pm k \lambda \\ \pm (2k+1) \frac{\lambda}{2} \end{cases}$
- ③ 迈克尔逊干涉仪: $\Delta d = (n-1)d = N \cdot \frac{\lambda}{2}$

$$\textcircled{1}: x = \begin{cases} \pm k \frac{D}{d} \lambda & k=0, 1, 2, \dots \\ \pm (2k+1) \frac{D}{2d} \lambda & k=0, 1, 2, \dots \end{cases} \quad k=0 \text{ 为中央明纹}$$

$$\Delta x = x_{k+1} - x_k = \frac{D}{d} \lambda$$

e.g: $d=0.2 \text{ mm}$, $D=1 \text{ m}$ (1) 若第一到第四明纹距离为 7.5 mm , 求入射光波长。

$$\Delta x_{1-4} = x_4 - x_1 = 3 \frac{D}{d} \lambda = 7.5 \text{ mm} \Rightarrow \lambda = 500 \text{ nm}$$

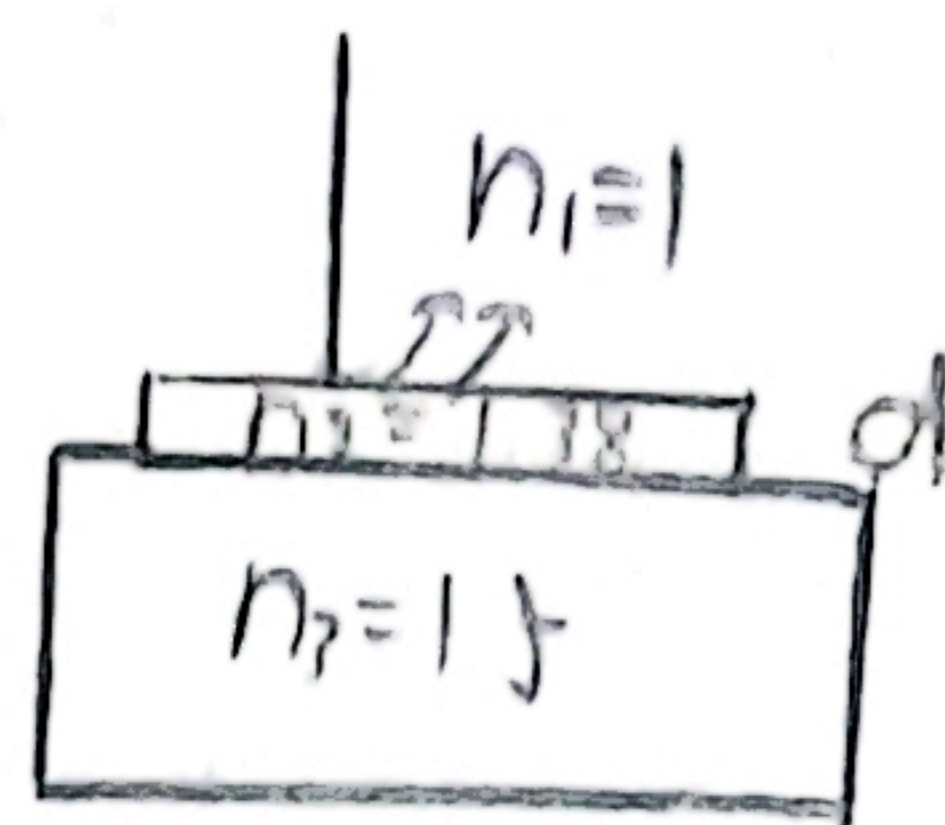
(2) 若 $\lambda = 600 \text{ nm}$, 求 Δx .

$$\Delta x = \frac{D}{d} \lambda = 3 \times 10^{-3} \text{ m} = 3 \text{ mm}$$

② 等倾干涉:

$$\delta(i) = 2d \sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2} = \begin{cases} \pm k\lambda \\ \pm (2k+1)\frac{\lambda}{2} \end{cases}$$

等倾条纹为一系列同心圆 - 内疏外密.



e.g 2: $\lambda = 550 \text{ nm}$, $n_3 = 1.5$, 其上涂有 $n_2 = 1.38 \text{ MgF}_2$ 增透膜. 光线垂直入射. 若反射光相消干涉中 $k=1$, 膜的厚度为多少? 在可见光范围内有无增反?

$$\delta = 2d \cdot n_2 = (2k+1)\frac{\lambda}{2} = \frac{3}{2}\lambda \Rightarrow d = 0.3 \mu\text{m}$$

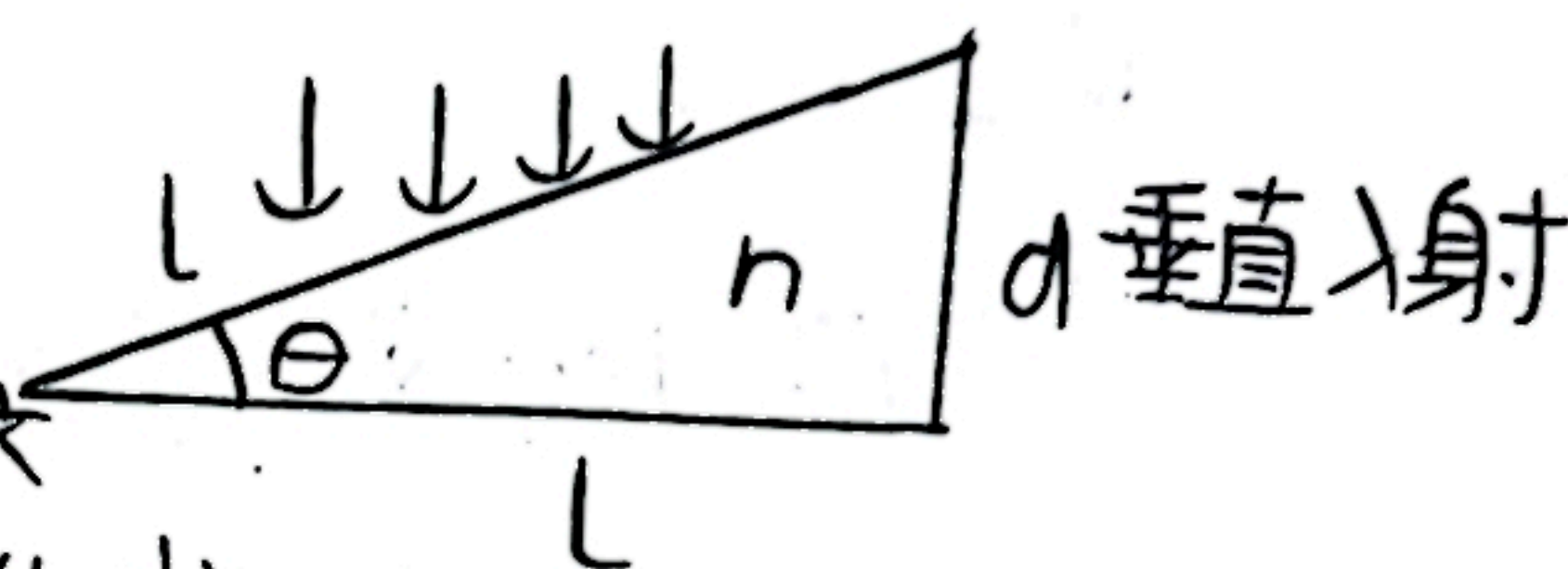
$$\text{增反: } \delta = 2n_2 d = k\lambda$$

$$k=1: \lambda_1 = 850 \text{ nm} > 760 \text{ nm} \quad k=2: \lambda_2 = 412.5 \text{ nm} > 400 \text{ nm (可见光)}$$

$$\text{e.g 3: } \delta = 2d \sqrt{n_2^2 - n_1^2 \sin^2 30^\circ} + \frac{\lambda}{2} = k\lambda \quad k=1 \Rightarrow d = 1.22 \times 10^{-7} \text{ m}$$

等厚干涉:

$$1. \text{劈尖: } \delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & \text{明} \\ (2k+1)\frac{\lambda}{2} & \text{暗} \end{cases}$$



$$\text{厚度差 } \Delta d = d_{k+1} - d_k = \frac{\lambda}{2n} = \frac{(k+1-\frac{1}{2})\lambda - (k-\frac{1}{2})\lambda}{2n}$$

$$\text{相邻间距 } l = \frac{\lambda}{2n \sin \theta} = \frac{\lambda}{2n\theta}$$

$$\text{劈尖公式: } \theta = \frac{\Delta d}{l} = \frac{\lambda}{2nl} \Rightarrow \lambda = 2nl\theta$$

$$2. \text{牛顿环: } \delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda \\ (2k+1)\frac{\lambda}{2} \end{cases}$$



$$R^2 - r^2 = (R-d)^2$$

$$r^2 = 2Rd \Rightarrow d = \frac{r^2}{2R}$$

$$r = \begin{cases} \sqrt{\frac{(2k-1)R\lambda}{2}} & \text{明} \\ \sqrt{kR\lambda} & \text{暗} \end{cases}$$

4. 光的衍射

① 夫琅禾费衍射: $\delta = a \sin \theta = \begin{cases} \pm 2k \frac{\lambda}{2} - \text{暗纹} \\ \pm (2k+1) \frac{\lambda}{2} - \text{明纹} \end{cases}$

半角宽度: $\Delta \theta_0 = \frac{\lambda}{a} (k=1)$ 角宽度: $2\Delta \theta_0 = \frac{2\lambda}{a}$ 线宽度: $l = 2\Delta \theta_0 f = 2f \frac{\lambda}{a}$

圆孔夫琅禾费衍射: $\sin \theta_1 = 1.22 \frac{\lambda}{a} = \theta_R$ 艾里斑半径: $R = f \cdot \tan \theta_1 = 1.22 \frac{\lambda}{a} f$

② 光栅衍射: 单缝衍射 + 多缝干涉

光栅常数 $d = a + b$ (透光宽度 + 不透光宽度) = $\frac{1}{N}$ (系数)

光栅方程: $a \sin \varphi = (a+b) \sin \varphi = \pm k \lambda$ (明纹)

缺级: $(a+b) \sin \varphi = k \lambda$ $a \sin \varphi = k' \lambda$ = 主极大

$\therefore k = \frac{a+b}{a} k'$

5. 光的偏振:

① 马吕斯定律: $I = I_0 \cos^2 \alpha$

② 布儒斯特定律: $\tan i_0 = \frac{n_2}{n_1}$

12.28-12.29

一. 静电场

1. 电荷. 库仑定律. 叠加定理 选择题
2. 电场与电场强度. 静止点电荷的叠加 填空题
3. 电场线与电通量. 高斯定理 选择题
4. 电势 选择题 / 填空题
5. 静电场中导体 判断题
6. 电介质

2. 电场强度: $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \vec{e}_r = \frac{F}{q_1} = \frac{kq_1q_2}{r^2} / q_1$

① 均匀无限长带电直棒: $E = \frac{\lambda}{2\pi\epsilon_0 a}$

② 带电圆环: $E = \frac{q_x}{4\pi\epsilon_0(x^2+R^2)^{3/2}}$ $\xrightarrow{x \gg R} E = \frac{q}{4\pi\epsilon_0 x^2}$

③ 带电圆盘: $E = \frac{\sigma}{2\epsilon_0} (1 - \frac{x}{\sqrt{R^2+x^2}})$ $\xrightarrow{R \gg x, \text{无限大}} E = \frac{\sigma}{2\epsilon_0}$

3. 高斯定理:

① 球面对称电场:

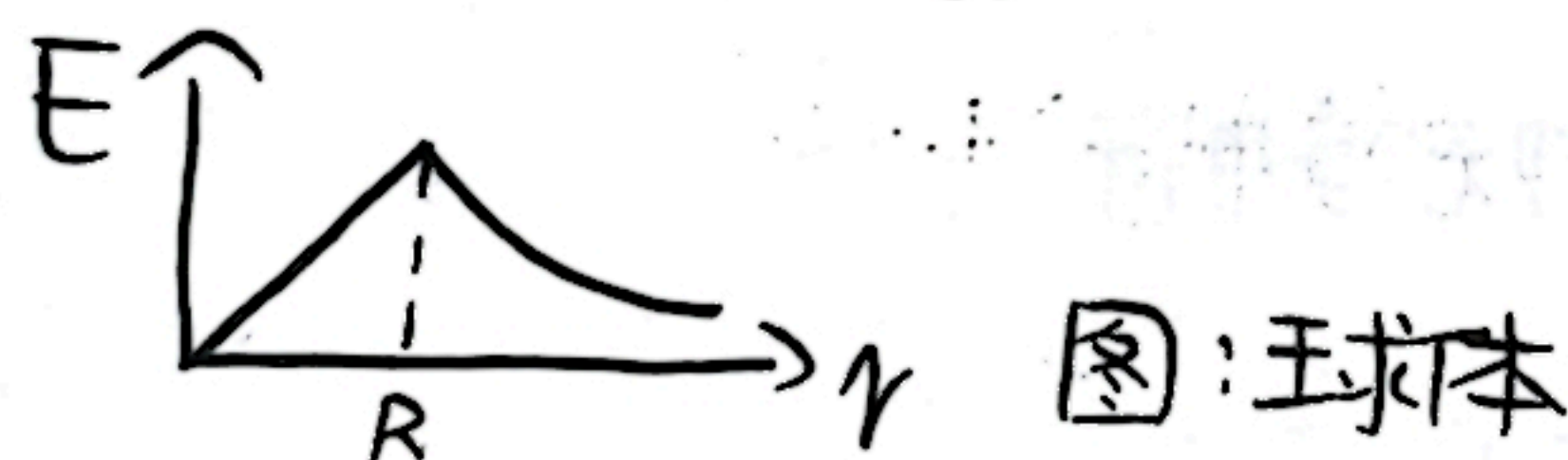
(1) q 均匀分布于球体内: I. P 点在球外: $\Psi E = 4\pi r^2 E = \frac{q}{\epsilon_0}$

$\therefore E = \frac{q}{4\pi\epsilon_0 r^2} \vec{e}_r$

II. P 点在球内: $\Psi E = \oint E dS = 4\pi r^2 E$ 高斯面 $q'(r < R) = \rho \cdot \frac{4}{3}\pi r^3$

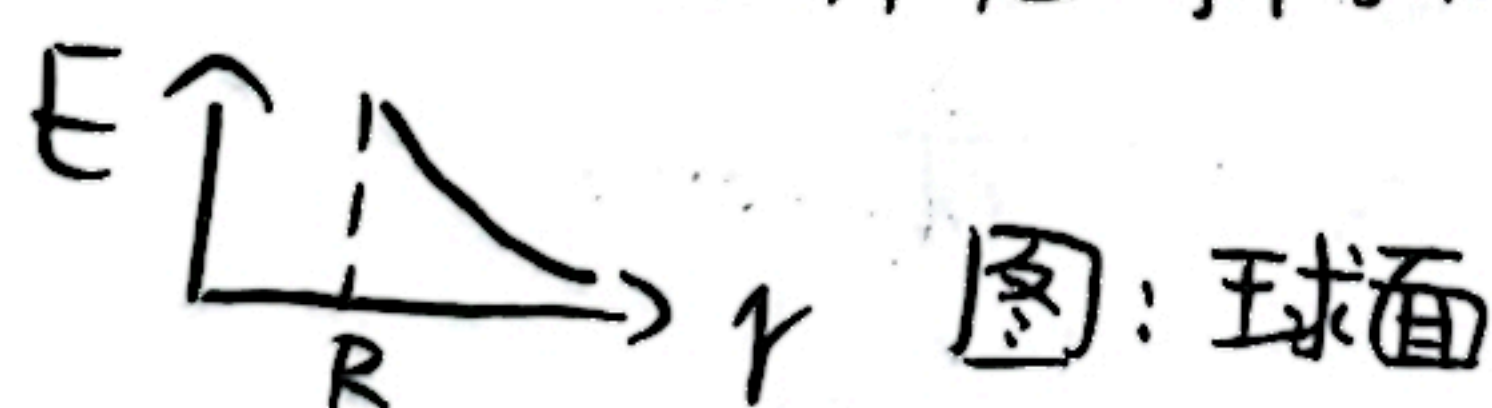
$4\pi r^2 E = \frac{q'}{\epsilon_0} = \frac{r^3}{R^3} q$

$E = \frac{qr}{4\pi\epsilon_0 R^3} \vec{e}_r = \frac{qr}{4\pi\epsilon_0 R^3} \vec{e}_r$ (随 r 线性增加)

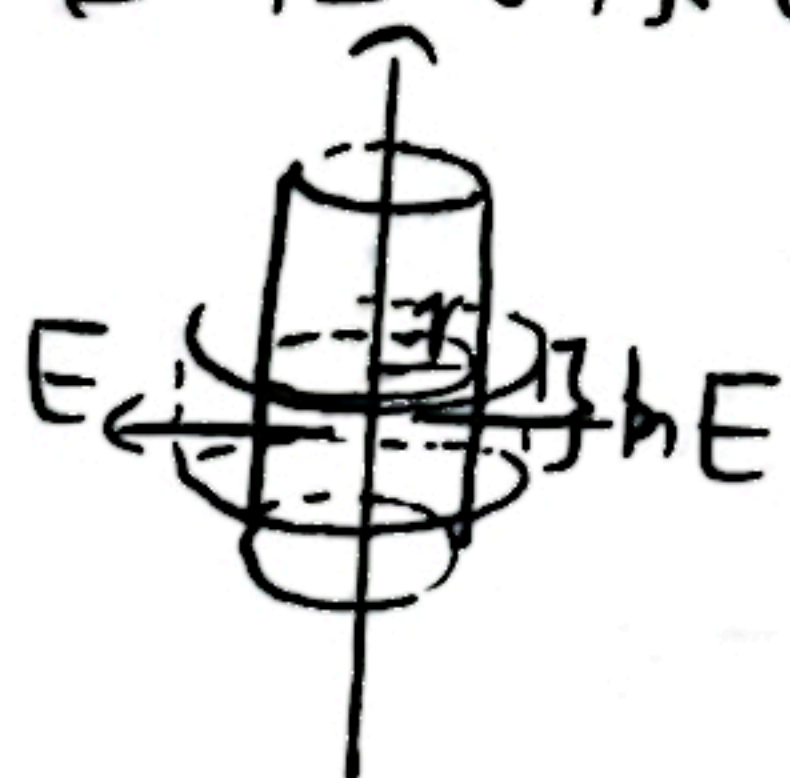


(2) q 均匀分布于球面上: I. P 在球外: $4\pi r^2 E = \frac{q}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi r^2 \epsilon_0}$

II. P 在球内: $q' = 0 \Rightarrow E = 0$



② 柱对称电场: λ (线密度) R (圆柱半径)



$\Psi E = 2\pi r h E$

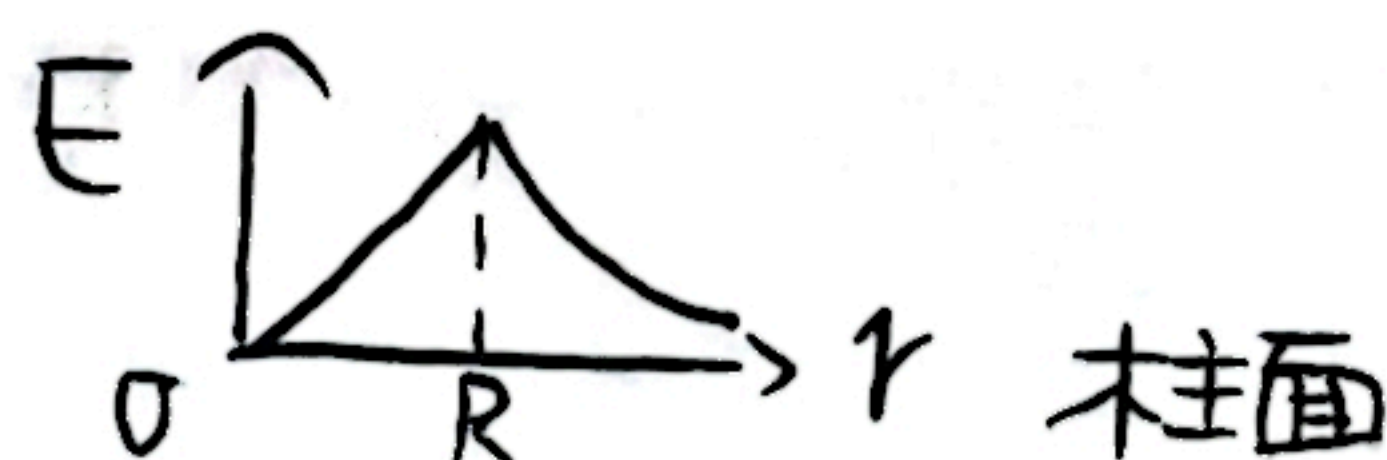
I. P 在圆柱外: $2\pi r h E = \frac{q}{\epsilon_0} = \frac{\lambda h}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi r \epsilon_0} \vec{e}_r$ (无限长带电直棒)

II. P 在圆柱内: $2\pi r h E = \frac{q'}{\epsilon_0}$

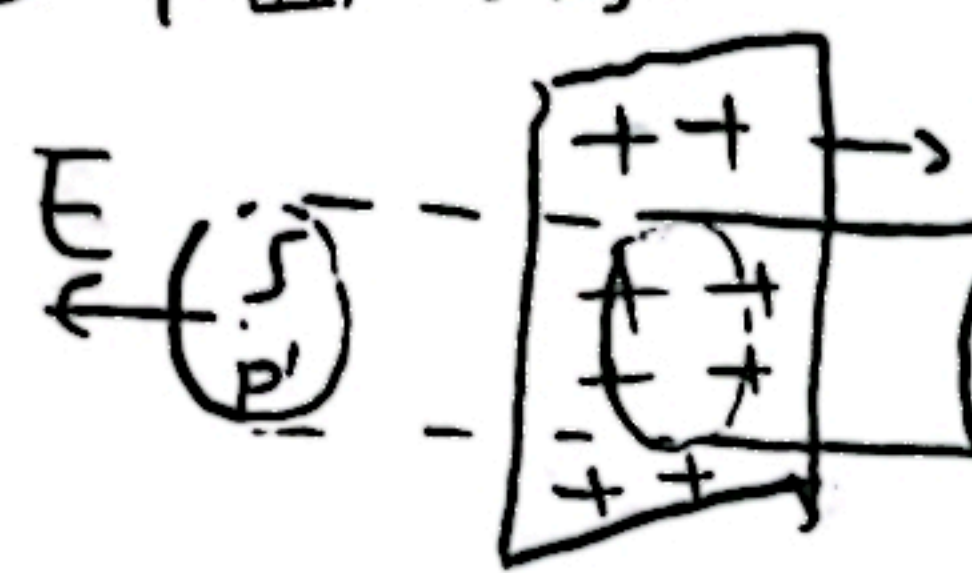
$\therefore 2\pi r h E = \frac{q}{\epsilon_0} \frac{r^2}{R^2} \Rightarrow E = \frac{qr}{2\pi\epsilon_0 R^2} \vec{e}_r$



$q = \pi R^2 \lambda h$
 $q' = \pi r^2 \lambda h$
 $\therefore q' = q \cdot \frac{r^2}{R^2}$



③ 平面电容器:



侧面: $E \perp$ 侧面 $\Psi_E = 0$ 面密度 $\frac{65(9)}{\epsilon_0}$
 底面: $\Psi_E = ES + ES = \frac{65(9)}{\epsilon_0}$
 $E = \frac{6}{2\epsilon_0}$ (均匀场)

4. 电势:

① 做功与环路定理:

$$A_{ab} = \frac{qq_0}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dr = \frac{qq_0}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right) = q_0 \int_a^b E dl \Rightarrow \frac{W}{q_0} = V$$

$$\oint E dl = 0 \Rightarrow \text{有源无旋}$$

$$W = q_0(V_a - V_b) = q_0 U_{ab}$$

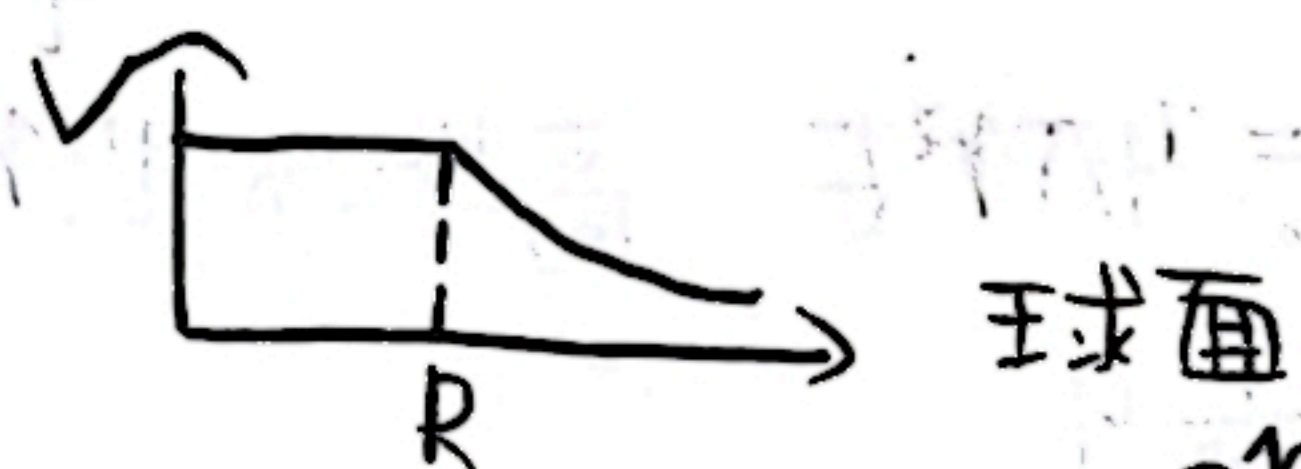
② 电势: $V_P = \frac{W}{q_0} = \frac{q}{4\pi\epsilon_0 r}$

(1) 带电球面的电势:

$$E = \begin{cases} 0 & r < R \\ \frac{q}{4\pi\epsilon_0 r^2} & r > R \end{cases}$$

$$r > R: V = \int_r^\infty \frac{q}{4\pi\epsilon_0 r'^2} dr' = \frac{q}{4\pi\epsilon_0 r}$$

$$r < R: V = \int_r^R E dr + \int_R^\infty E dr = \frac{q}{4\pi\epsilon_0 R}$$



(2) 无限长带电直线: $-\Delta V = U = \int_r^{r'} E dl = \frac{\lambda}{2\pi\epsilon_0} \int_r^{r'} \frac{1}{r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r'}{r}$

5. 静电场中导体:

① 静电平衡

- 导体内场强为0: 内部无净电荷
- 导体是等势体
- 导体表面电场强度垂直于导体表面

② 电容: $C = \frac{q}{V}$

(1) 平行板电容器: $E = \frac{6}{2\epsilon_0} + \frac{6}{2\epsilon_0} = \frac{6}{\epsilon_0}$ $U = Ed = \frac{6}{\epsilon_0} d$

$$C = \frac{q}{U} = \frac{65}{\frac{6d}{\epsilon_0}} = \epsilon_0 \frac{S}{d}$$

(2) 圆柱形电容器: $E = \frac{\lambda}{2\pi\epsilon_0 r}$ $U_{AB} = \int_{R_A}^{R_B} E dl = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_B}{R_A}$

$$C = \frac{Q}{U_{AB}} = \frac{\lambda L}{\frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_B}{R_A}} = \frac{2\pi\epsilon_0 L}{\ln \frac{R_B}{R_A}}$$

(3) 球形电容器: $E = \frac{q}{4\pi\epsilon_0 r^2}$ $U_{AB} = \int_{R_A}^{R_B} E dl = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_A} - \frac{1}{R_B} \right)$

取 $R_B \rightarrow \infty$ $C = \frac{q}{U} = 4\pi\epsilon_0 R_A$

(4) 串联: $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots = \sum_{i=1}^n \frac{1}{C_i}$ (Q恒定) 并联: $C = C_1 + C_2 = \sum_{i=1}^n C_i$ (U相同)

充电介质: $C = \epsilon_r C_0$ ($\epsilon_r > 1$)

12.29

二. 稳恒磁场

- | | |
|--------------|----|
| 1. 恒定电流 | 选择 |
| 2. 磁场强度 | 选择 |
| 3. 毕奥萨伐尔定律 | 填空 |
| 4. 安培环路定理 | 填空 |
| 5. 磁场对运动电荷作用 | 选择 |
| 6. 磁介质 | 判断 |

1. 恒定电流: 大小和方向不随时间而变化. (标量)

电流密度: $j = \frac{dI}{dS_\perp}$ (矢量) $I = \int_S j \cdot dS$

电动势 $\mathcal{E} = \frac{dA}{dq} = \oint_L E_k \cdot dl$ (有旋)

2. 磁场强度: $\oint H \cdot dl = \sum I$ $B = \mu H = \mu_0 \mu_r H$

3. 毕奥萨伐尔定律: $dB = \frac{\mu_0}{4\pi} \frac{Idl \times \vec{e}_r}{r^2}$



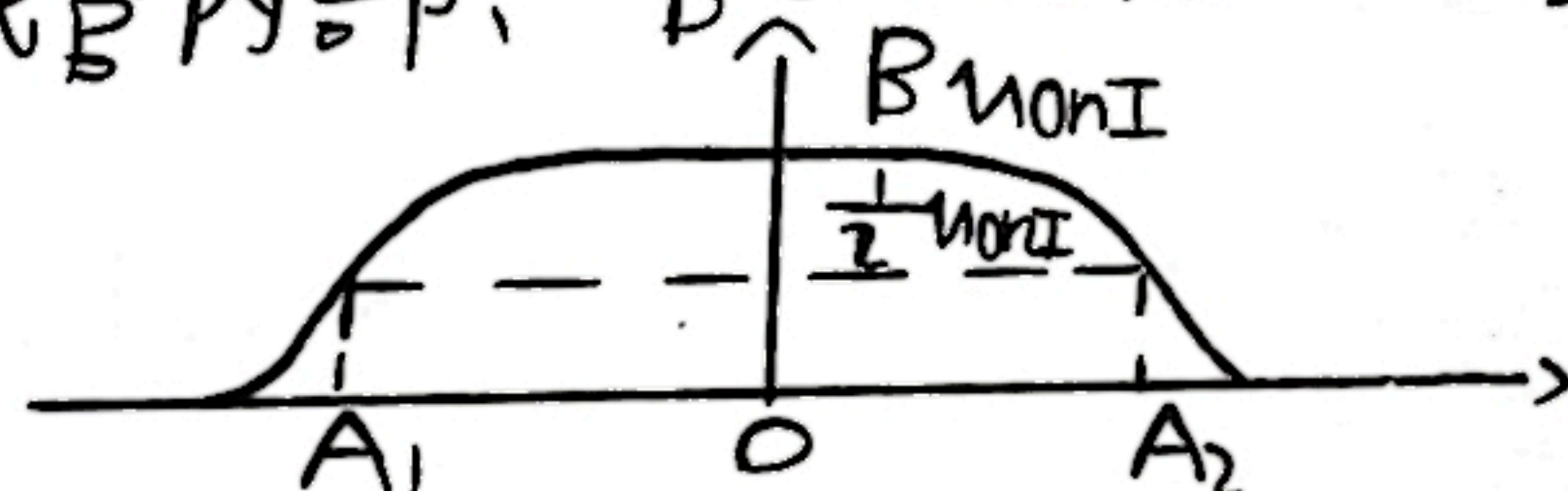
① 无限长直导线: $B = \frac{\mu_0 I}{4\pi a} (\sin \theta_1 - \sin \theta_2)$ $\frac{\theta_1 = \frac{\pi}{2}}{\theta_2 = -\frac{\pi}{2}} \Rightarrow B = \frac{\mu_0 I}{4\pi a} (\cos \theta_1 - \cos \theta_2)$

② 载流圆线圈: $B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{\frac{3}{2}}} = \frac{\mu_0 IS}{2\pi(R^2 + x^2)^{\frac{3}{2}}}$

(1) 在圆心 $x=0$ 处: $B = \frac{\mu_0 I}{2R}$

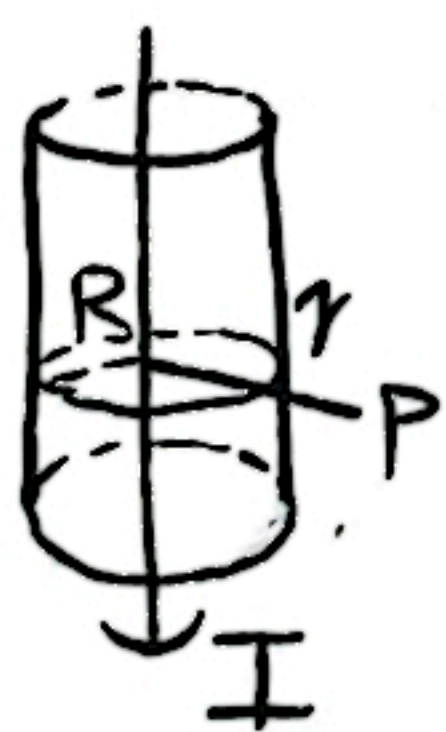
(2) $x \gg R$: 令 $m = IS$ $B = \frac{\mu_0 IS}{2\pi x^3} = \frac{\mu_0 m}{2\pi x^3}$

③ 载流直螺线管内部: $B = \mu_0 n I$ 端点: $B = \frac{1}{2} \mu_0 n I$



4. 安培环路定理: $\oint B \cdot dl = \mu_0 \sum I$

① 长直圆柱形载流导线:



(1) P 在外: $B \cdot 2\pi r = \mu_0 I$ ($r > R$)

$B = \frac{\mu_0 I}{2\pi r}$

(2) P 在内: $B \cdot 2\pi r = \mu_0 I' = \mu_0 \frac{\pi r^2}{\pi R^2} I$ ($r < R$) $(\frac{I'}{I} = \frac{\pi r^2}{\pi R^2} = \frac{r^2}{R^2})$

$\therefore B = \frac{\mu_0 I}{2\pi R^2} r$

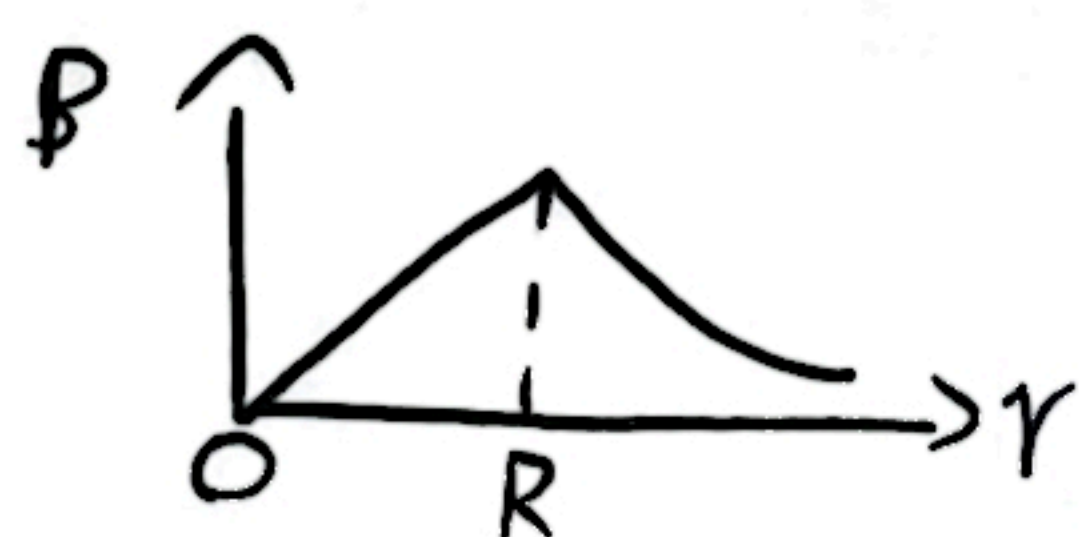
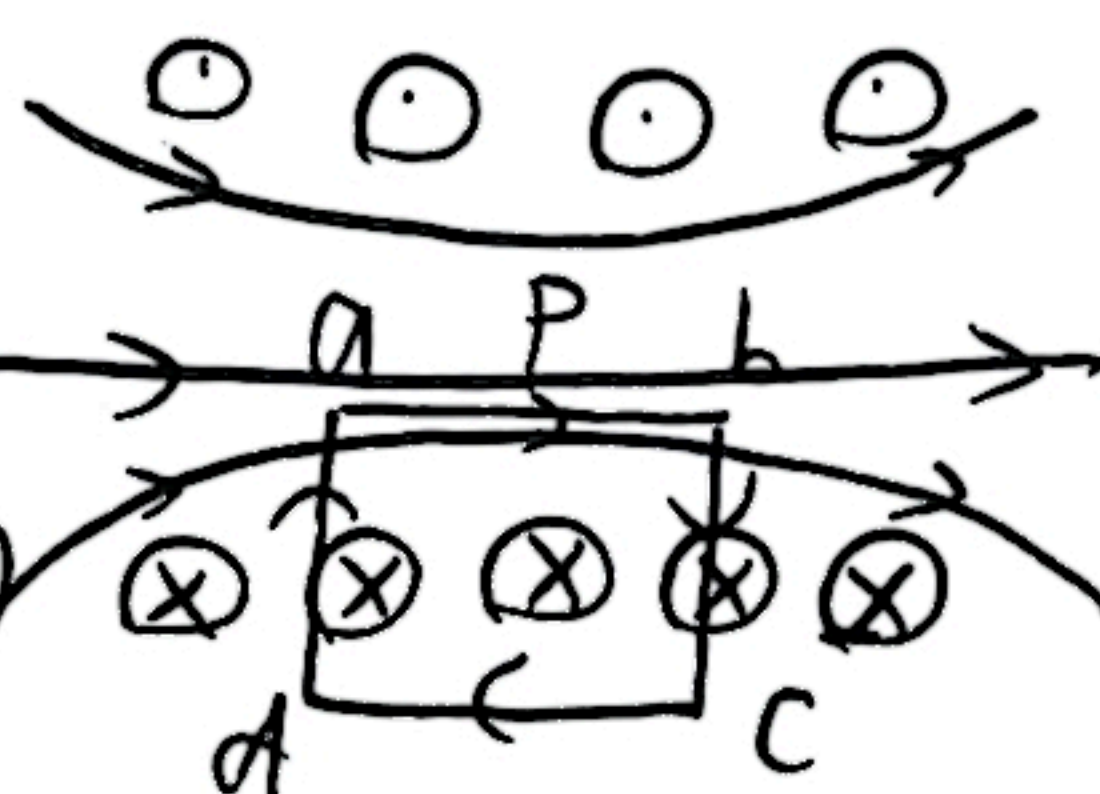


图: 圆柱

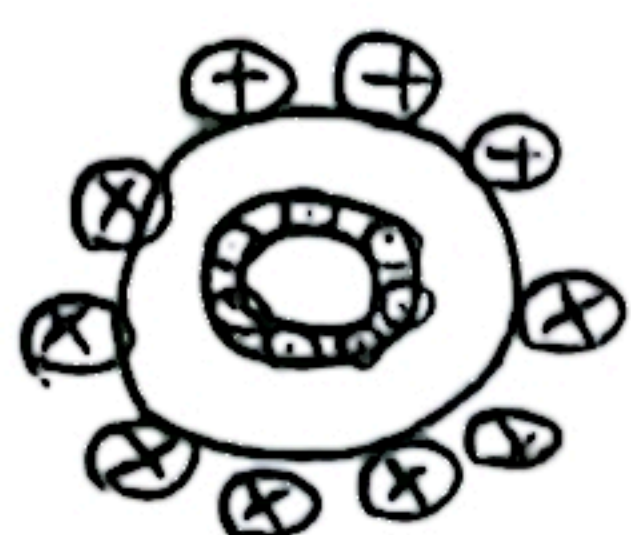


② 载流长直螺线管内:

$\oint B \cdot dl = B \cdot |ab|$ (其余为0)
 $= \mu_0 n I |ab|$

$B = \mu_0 n I = \frac{\mu_0 N I}{L}$

③ 载流螺绕环



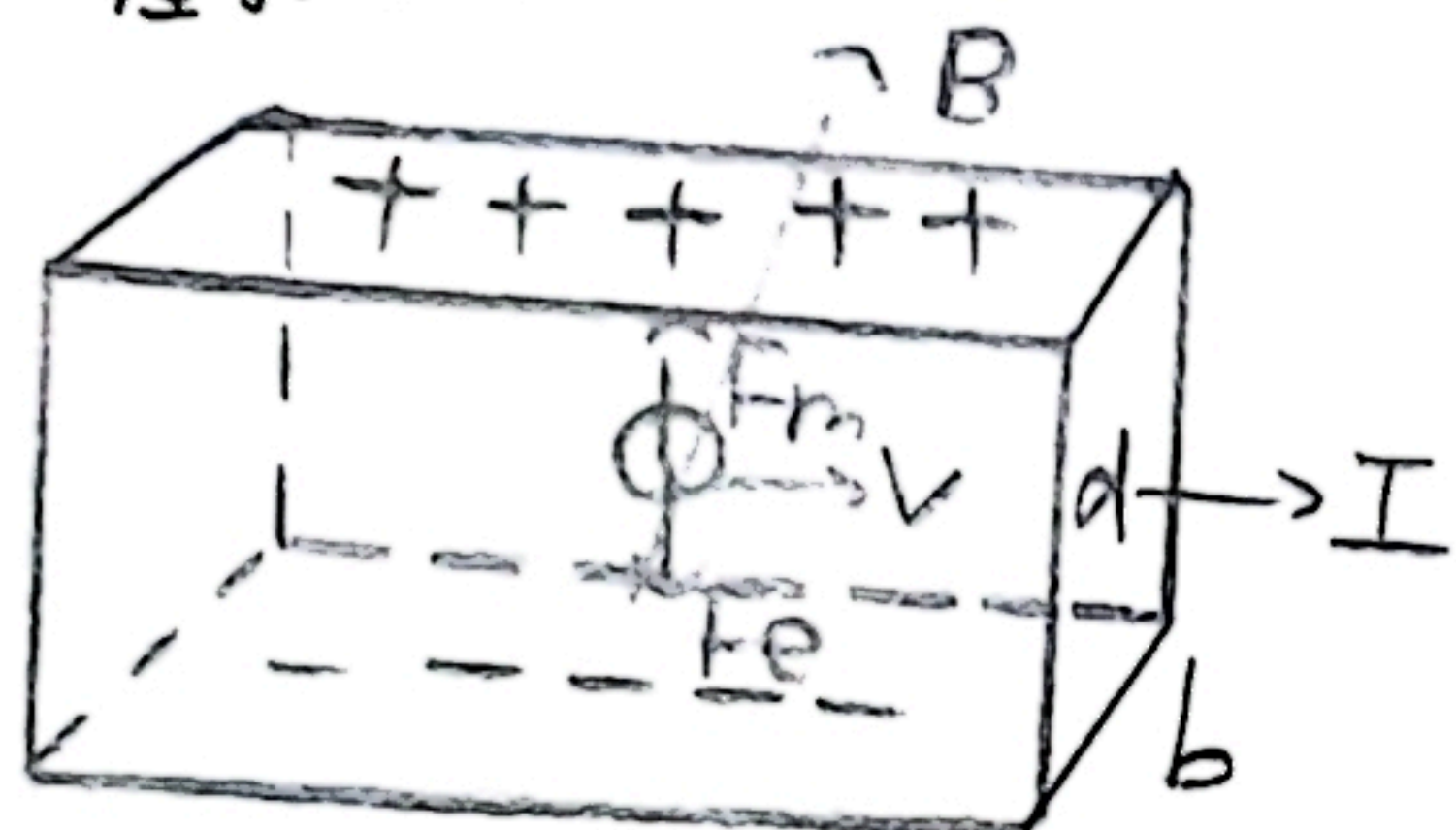
$$\oint B dl = B \cdot 2\pi r = \mu_0 NI \Rightarrow B = \frac{\mu_0 NI}{2\pi r} = \frac{\mu_0 NI}{L} = \mu_0 n I$$

5. ① $R = \frac{m v_0}{q B}$

螺旋线: $R = \frac{m v_{\perp}}{q B} = \frac{m v_0}{q B} \sin \theta$

螺距: $h = v_{\parallel} T = \frac{2\pi m}{q B} v_0 \cos \theta$

② 霍尔效应:



$$\begin{cases} qE = qvB \Rightarrow E = vB \\ U = Ed \\ I = nesv = nebdv \end{cases} \Rightarrow \begin{cases} E = BV \\ U = Bvd = \frac{1}{ne} \cdot \frac{IB}{b} = R_H \frac{IB}{b} \end{cases}$$

6. $B = B_0 + B'$

磁性物质

- 顺磁质: $B > B_0$ (稍大于)
- 抗磁质: $B < B_0$ (稍小于)
- 铁磁质: $B \gg B_0$ (远大于)

12.31 电磁感应

1. 电磁感应定律 选择
2. 动生电动势 填空
3. 感生电动势, 感生电场 简答+判断
4. 自感与互感 选择
5. 磁场的能量 判断

$$1. \quad \mathcal{E}_i = -N \frac{d\Phi}{dt} = \frac{-dN\Phi}{dt} \rightarrow \text{磁链} \quad I_i = -\frac{1}{R} \frac{d\Phi}{dt}$$

$$q = \int_{t_1}^{t_2} I_i dt = -\frac{1}{R} \int_{t_1}^{t_2} \frac{d\Phi}{dt} dt = -\frac{1}{R} \int_{\Phi_1}^{\Phi_2} d\Phi = \frac{1}{R} (\Phi_1 - \Phi_2)$$

$$\mathcal{E}_i = \oint_L \mathbf{E}_k \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}$$

$$2. \quad \mathcal{E}_i = \int \mathbf{v} \times \mathbf{B} \cdot d\mathbf{l}$$

① 铜棒:



$$\mathcal{E}_i = \int \mathbf{v} \times \mathbf{B} \cdot d\mathbf{l} = \int_0^L B v \cdot dl = \frac{B v L^2}{2}$$

方向 $A \rightarrow O$ $v_O > v_A$

② 导线:



$$\begin{aligned} \mathcal{E}_i &= \int \mathbf{v} \times \mathbf{B} \cdot d\mathbf{l} = \int v \cdot \frac{\mu_0 I}{2\pi x} dx \\ &= \frac{\mu_0 I}{2\pi} v \int_a^{a+L} \frac{dx}{x} = \frac{\mu_0 I}{2\pi} v \ln \frac{a+L}{a} \end{aligned}$$

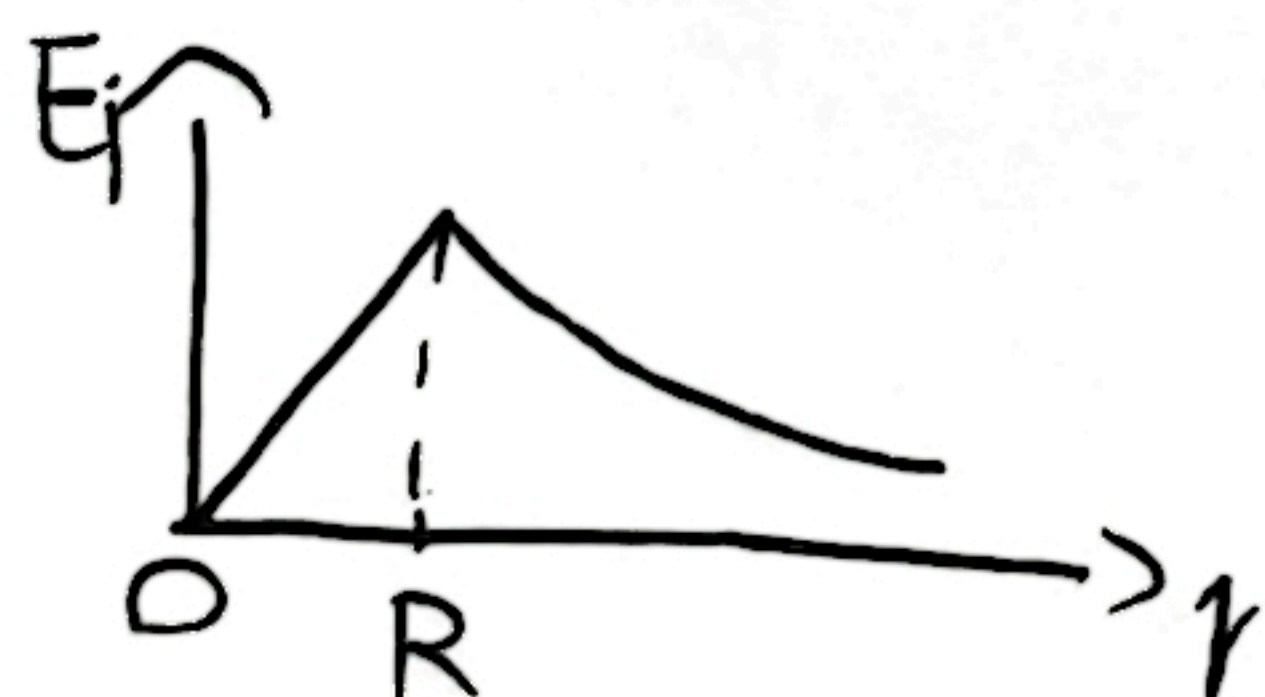
$v_M > v_N$

$$3. \quad \oint_L \mathbf{E}_i \cdot d\mathbf{l} = \mathcal{E}_i = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \quad (\text{涡旋电场})$$

① 长螺线管: (1) 管内: $\oint_L \mathbf{E}_k \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \quad (r < R)$

$$E_k \cdot 2\pi r = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = -\pi r^2 \frac{dB}{dt}$$

$$E_k = -\frac{r}{2} \frac{dB}{dt} = -\frac{r}{2} k$$



E-r 图

(2) 管外: $2\pi r \cdot E_k = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = -\pi R^2 \frac{dB}{dt}$

$$E_k = \frac{-R^2}{2r} \frac{dB}{dt}$$

② 圆柱:



(1) $\mathcal{E}_i = \int_a^b \mathbf{E}_i \cdot d\mathbf{l} = \int_a^b \frac{r}{2} \frac{\partial B}{\partial t} \cos\theta \cdot dl$

$$\cos\theta = \frac{\sqrt{R^2 - (r/2)^2}}{r}$$

$$\therefore \mathcal{E}_i = \frac{1}{2} \frac{dB}{dt} \int_a^b r \frac{\sqrt{R^2 - (r/2)^2}}{r} dl = \frac{L}{2} \sqrt{R^2 - (r/2)^2} \frac{dB}{dt}$$

(2) $\mathcal{E}_i = -\frac{d\Phi}{dt} = -\frac{d}{dt} [-B \frac{L}{2} \sqrt{R^2 - (r/2)^2}] = \frac{L}{2} \sqrt{R^2 - (r/2)^2} \frac{dB}{dt}$

简答: 涡电流 (释放大量焦耳热)

$$4. \quad \varepsilon_L = -L \frac{dI}{dt} \quad L = \frac{d\Phi_N}{dI} = \frac{\Phi_N}{I} \quad (\text{自感})$$

$$\varepsilon_1 = -M \frac{dI_2}{dt} \quad \varepsilon_2 = -M \frac{dI_1}{dt} \quad (\text{互感}) \quad M = \frac{\Phi_{21}}{I_1} = \frac{\Phi_{12}}{I_2} = k\sqrt{L_1 L_2}$$

$$5. \quad W_m = \frac{1}{2} L I_0^2 \quad w_m = \frac{1}{2} B \cdot H$$

